Errata for *A Time and Space Optimal Stable Population Protocol Solving Exact Majority*

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found while formalizing the paper in Lean 4 / Mathlib

Abstract

We formalized Theorem 3.1 of the paper in Lean 4 with Mathlib: the protocol NONUNIFORM MAJORITY uses $O(\log n)$ states, stably computes majority, and stabilizes in $O(\log n)$ parallel time with probability $1 - O(1/n^2)$ and in expectation. Theorem 3.1 holds. Along the way we found the items below. Two printed statements are false as stated: the first case of Lemma 5.3 for large gaps, and the lower bound of Theorem 6.9 for small clock fractions, which comes from a wrong conversion factor. Neither affects Theorem 3.1. The first needs one extra step in the expected-time argument; the second leaves the choice $k = 45$ in Theorem 6.12 without a proof, and a larger constant k restores it. The other items are proof gaps with true statements, and typographical or pseudocode slips whose intended meaning is clear from the text.

All locations refer to arXiv:2106.10201v2; page numbers are those of its PDF. Classification: (E) a printed statement is false; (P) the statement is true but the printed proof does not establish it; (T) a typographical or pseudocode slip with unambiguous intended meaning.

Notation

As in the paper, n is the population size, g the initial gap, M the set of Main agents, $c = |C|/n$ the fraction of Clock agents, $m = |M|/n$, and k the number of minutes per hour. Parallel time is the number of interactions divided by n.

Erratum 1 (Lemma 5.3, large gaps (E)). Location. Lemma 5.3 and its proof (p. 26); the description of the Phase 2 Init (p. 18); the expected-time paragraph of the proof of Theorem 3.1 (p. 46).

As printed. "If the initial gap $|g| \geq 0.025|M|$, then we stabilize to the correct output in Phase 2" with probability $1 - O(1/n^2)$. The proof treats $|g| > 0.5|M|$ by noting $|\mu| \geq 1$, "so all remaining biased agents have the majority opinion, and we will stabilize in Phase 2 to the correct majority output." On p. 18 the check $|\text{bias}| > 1$ in the Phase 2 Init "can only happen with low probability."

Issue. Phases 0 and 1 conserve the sum of the biases of Main and RoleMCR agents, which equals g; a RoleMCR agent that reaches Phase 1 sends the population to Phase 10. Main agents

leave Phase 1 only through lines 1–4 of the main loop, which execute the Phase 2 Init, and no Phase 2 rule changes bias. If $|g| > |M|$, consider the last Main agent to leave Phase 1. Every other Main agent left with $|\text{bias}| \le 1$ or has already triggered the error, so this agent holds bias of absolute value at least $|g| - (|M| - 1) \ge 2$. Its Phase 2 Init sends it to Phase 10 (reading the main loop as in Erratum 9), and the epidemic brings the whole population to the backup. Thus for every input with $|g| > |M|$ the protocol reaches Phase 10 with probability 1 once Phase 1 has started. Since $|M| \le 2n/3$ at all times (from the invariant $s_f + 2s_t = m_f + 2m_t$ in the proof of Lemma 5.2; the lemma states the bound under the condition that Phase 1 initializes without error), this includes every input with $|g| > 2n/3$, in particular the unanimous input $g = n$. The range $0.5|M| < |g| \le |M|$ has a related step to check: rounding gives $\mu = 1$, and the cited convergence to $\{\mu - 1, \mu, \mu + 1\} = \{0, 1, 2\}$ does not by itself exclude an agent with bias 2.

Correction. The output remains correct, since Phase 10 is a stable backup. For the time bound, the expected-time paragraph on p. 46 charges Phase 10 only on an event of probability $O(1/n^2)$ and uses the $O(n \log n)$ bound of Lemma 7.7 there. For large gaps the backup is fast. Suppose some agent is in Phase 10 and $|g| \ge n/120$ (which holds whenever $|g| \ge 0.025|M|$, since $|M| \ge n/3$). The phase epidemic to Phase 10 completes in $O(\log n)$ parallel time. In an all-Phase-10 configuration the number of active majority agents exceeds the number of active minority agents by $|g|$, and the backup stabilizes after three monotone stages (eliminate the active minority, then the active ties, then correct the passive outputs), each with useful-partner probability at least $2|g|/(n(n-1))$ per target agent. This gives expected stabilization within $180(1 + \ln n)$ parallel time from an all-Phase-10 configuration, and $O(\log n)$ parallel time with probability $1 - O(n^{-3})$. With this step, Theorem 3.1 holds for all inputs.

Erratum 2 (Theorem 6.9, conversion to the whole population (E)). Location. Theorem 6.9 and its proof (p. 33); its use with $k = 45$ in the proof of Theorem 6.12 (p. 35) and in the proof of Lemma 6.14 (p. 36).

As printed. The lemmas of §6.1 assume $|C| = n$. The proof of Theorem 6.9 then multiplies their parallel-time bounds by $1/c^2$, since two Clock agents meet with probability about c^2, and obtains $\text{end}_h - \text{start}_h \ge \frac{1}{c^2}[0.45k - 3.1]$ and $\text{start}_{h+1} - \text{start}_h \le \frac{1}{c^2}[2.11k + 2.2]$.

Issue. The factor $1/c^2$ converts a number of Clock–Clock interactions into a number of global interactions. The parallel time of the all-clock lemmas, however, is measured in units of $|C|$ interactions, while global parallel time is measured in units of n. The correct factor is

$$\frac{|C|}{n} \cdot \frac{n(n-1)}{|C|(|C|-1)} = \frac{n-1}{|C|-1} \sim \frac{1}{c}.$$

The printed upper bound remains valid (it is weaker than the corrected one, $\frac{1}{c}[2.11k + 2.2]$). The lower bound on the length of a synchronous hour becomes about $\frac{1}{c}[0.45k - 3.1]$, smaller by a factor of c. Since $\text{end}_h \le \text{start}_{h+1}$, the printed lower bound exceeds the corrected upper bound when $c < (0.45k - 3.1)/(2.11k + 2.2)$ (about 0.18 for $k = 45$), so as a statement about an arbitrary clock fraction c it does not hold.

Effect. Theorem 6.12 needs each synchronous hour to last at least $\frac{2}{c} + \frac{47}{m} \le \frac{49}{c}$ (using $c < m$). It uses $\frac{0.45 \cdot 45 - 3.1}{c^2} \ge \frac{17}{c^2} \ge \frac{49}{c}$. With the corrected factor, $k = 45$ gives $\frac{17.15}{c} < \frac{49}{c}$, so the argument no longer applies; we do not claim that the conclusion fails. The proof of Lemma 6.14 uses $[1.48(45-2) - 2.2]/c^2 \ge 17/c^2 \ge 51/m$ in the same way. (The factor 1.48 there should be 0.45, the per-minute lower bound for $p = 1$; the printed chain is unaffected, since $0.45 \cdot 43 - 2.2 = 17.15$.)

Correction. Take a larger constant number of minutes per hour. Theorem 6.12 needs $0.45k - 3.1 \geq 49$, that is $k \geq 116$, and Lemma 6.14 needs $0.45k - 3.1 \geq 51$, that is $k \geq 121$. With a 5% allowance for the concentration factors, $k = 128$ satisfies both: $0.95(0.45 \cdot 128 - 3.1) = 51.775$. This changes only a constant; the state count stays $O(\log n)$ and the asymptotic statements are unaffected. Our formal proof uses $k = 256$, which leaves more room in the later phases.

Erratum 3 (Lemma 6.7, the displayed integral (T)). Location. Proof of Lemma 6.7 (p. 31), last display.

As printed.

$$c_{\geq i+1}(t_i^{0.1} + 0.45) \lesssim (1+\epsilon)e^{0.9}\left(0.01 + \int_0^{0.45} (1+s)^2 e^{-2s}\, ds\right) \leq 0.45.$$

Issue. The preceding text bounds the drip probability by $(0.1+s)^2$, so the integrand should be $(0.1+s)^2 e^{-2s}$. With $(1+s)^2$ the value is about 1.07. Also, the lemma needs $c_{\geq i+1}(t_i^{0.1}+0.45) < 0.1$, which the bound 0.45 does not give. (The sum before the integral counts agents rather than fractions; a factor $1/n$ is missing there.)

Correction. With the intended integrand,

$$e^{0.9}\left(0.01 + \int_0^{0.45} (0.1+s)^2 e^{-2s}\, ds\right) = 0.315\, e^{0.9} - 0.67625 \approx 0.0985 < 0.1,$$

so the lemma holds for ϵ small enough.

Erratum 4 (Section 6, asynchronous entry into Phase 3 (P)). Location. §3.4 (p. 11): "To simplify the analysis, we formally start our arguments for each phase assuming each agent is in the current phase, although technically $\Theta(\log n)$ time will pass between the time the first agent enters phase i and the last agent does"; its use in §6 (the clock lemmas of §6.1 and the induction on synchronous hours in §6.2).

Issue. The paper states this simplification explicitly. For Phase 3 it needs an argument. The analysis of §6 starts from a configuration in which all Clock agents are at minute 0 and the whole population is in Phase 3 (the base case $c_{\geq 0}(0) = 1$ of Theorem 6.5, $\Phi(0) = 0$ in Lemma 6.10, the hour-0 case of the induction). In the protocol, agents enter Phase 3 by epidemic over $\Theta(\log n)$ parallel time, so the first Clock agents drip and drag O agents forward while other agents are still in Phase 2 (or Phase 1). By the time the last agents enter, the clock can be $\Theta(\log n)$ minutes ahead. A configuration in which the whole population has just entered Phase 3 with all minutes at 0 occurs only with probability $e^{-\Omega(n)}$, so the analysis cannot be restarted there.

Correction. The clock argument has to start at the first entry into Phase 3, a stopping time, using the strong Markov property, and account for the width of the entry front. We did this in our formal proof. The conclusions used later (synchronous hours of length $\Theta(1/c)$, and the output of Theorems 6.1 and 6.2 at the end of Phase 3) hold with changed constants. The same treatment is needed at every phase boundary where the paper says "before the first Clock agent advances to the next phase."

Erratum 5 (Lemma 6.10, time horizon (P)). Location. Proof of Lemma 6.10 (pp. 33–34).

As printed. Azuma's inequality is applied to Φ over $n \cdot \mathrm{end}_h$ steps, giving $\exp(-O(n)\delta^2)$, "since by Theorem 6.9, we have time $\mathrm{end}_h = O(1)$."

Issue. Time is measured from the start of Phase 3, so end_h is $\Theta(h)$, which is $\Theta(\log n)$ for the last hours. The horizon is then $O(n \log n)$ steps and the bound is $\exp(-\Omega(n/\log n))$.

Correction. This is still a very high probability bound, and the union over hours goes through.

Erratum 6 (Hour of a Clock agent, floor or ceiling (T)). Location. §3.4, Phase 3 description (p. 18) and pseudocode (p. 19); the definition of a.hour (p. 32); proof of Theorem 6.9 (p. 33).

Issue. The text on p. 18 defines hour $= \lceil \text{minute}/k \rceil$, while the clock update reaction uses $\lfloor c.\text{minute}/k \rfloor$, p. 32 defines $a.\text{hour} = \lfloor a.\text{minute}/k \rfloor$, and the proof of Theorem 6.9 uses $c.\text{hour} = h \iff hk \le c.\text{minute} < (h+1)k$.

Correction. Read the floor throughout. With the floor, hour L corresponds to the single value minute $= kL$; this is harmless, since Clock agents at minute $= kL$ then wait $c_3 \ln n$ on their counter before Phase 3 ends.

Erratum 7 (Phase 7 pseudocode, gap-2 cancel (T)). Location. Phase 7 pseudocode, lines 7–10 (p. 20).

Issue. Executed in order, line 9 sets $i.\text{exponent} \leftarrow i.\text{exponent} - 1$ and line 10 then sets $j.\text{exponent} \leftarrow i.\text{exponent} - 2$ with the new value, so j ends three below the original exponent of i. The example in the comment, $+\frac{1}{4}, -\frac{1}{16} \to +\frac{1}{8}, +\frac{1}{16}$, and conservation of mass ($\frac{1}{4} - \frac{1}{16} = \frac{1}{8} + \frac{1}{16}$) require two below.

Correction. By the guard on line 7, j already has exponent two below the original exponent of i, so line 10 can be deleted (or executed before line 9).

Erratum 8 (Phase 8 guard (T)). Location. Phase 8 pseudocode, line 1 (p. 21).

Issue. The guard $\{u.\text{bias}, v.\text{bias}\} = \{-1, +1\}$ read literally requires both agents at exponent 0. The description of Phase 8 (p. 20) concerns "opposite-opinion agents of any two exponents," and line 2 compares exponents.

Correction. The guard should read $\{u.\text{opinion}, v.\text{opinion}\} = \{-1, +1\}$.

Erratum 9 (Main loop, phase update after an Init (T)). Location. NONUNIFORM MAJORITY, lines 1–4 (p. 12).

Issue. If one of the Inits executed in line 3 moves agent i to Phase 10 (the error checks in the Inits of Phase 1 and Phase 2), the assignment $i.\text{phase} \leftarrow j.\text{phase}$ in line 4 moves it back down, and the loop runs the later Inits. Elsewhere the paper treats the phase as nondecreasing (p. 11). Erratum 1 uses the intended reading.

Correction. Stop the loop once i reaches Phase 10, and set $i.\text{phase} \leftarrow \max(i.\text{phase}, j.\text{phase})$.

Erratum 10 (Small slips (T)).

1. p. 18: the O agents' field is introduced as "hour $= L$", while the Init sets hour $\leftarrow 0$; and "will only participate in split reactions with $-$exponent $>$ hour" is reversed relative to pseudocode line 13 and Fig. 4c.

2. p. 33: the definition of end_h uses "$\le 0.001|C|$" where "reaches 0.001" (that is, $\ge$) is meant; in the proof of Theorem 6.9, $\text{end}_h = t_{hk}^{0.001}$ should be indexed $(h+1)k$, and "$t_{i+1} - t_i \le 0.45/c^2$" should be $\ge$.

3. Proof of Lemma 7.5 (p. 45): in the step for $B_{-(l+1)}$, "Initially the count is at most $|B_{-(l+2)}| \leq 0.008|M|$" should read $|B_{-(l+1)}|$.

4. Proof of Theorem 3.1 (p. 46): "after Phase 7 and Phase 7" should read "after Phase 7 and Phase 8."

5. Proof of Theorem 3.1 (p. 46): "We finally justify that the expected stabilization time is $O(n \log n)$" should read $O(\log n)$, as concluded at the end of the paragraph.

6. Very small populations: Phase 0 ends only through Clock–Clock interactions, and for $n = 3$ (input A, A, B) it produces at most one Clock agent, so Phase 0 never ends. The sentence on p. 46, "By Lemma 5.2, we must produce at least 2 Clock agents by the end of Phase 0," needs a lower bound on n. This does not affect the asymptotic theorem.

Formal verification

The formalization is in the public repository `https://github.com/zinan-huang/Ripple`, directory `Ripple/PopulationProtocol/ExactMajority`. It follows the paper's protocol and its proof phase by phase, with the corrections above and the clock parameter $k = 256$. The main theorem is `TheoremThreeOne.exact_majority_theorem_three_one`. For every population size n above an explicit threshold it states, for one protocol: the number of states is at most a constant times $\lceil \log_2 n \rceil + 1$; the protocol stably computes majority; and for every valid input of size n the stabilization time exceeds a constant times $n \log n$ interactions with probability at most $110/n^2$, and its expectation is $O(n \log n)$ interactions, with explicit constants (not optimized). Correctness alone is proved for all $n \geq 11$. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.